Measure Theory

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1 Measure Theory (測度論)

I σ-algebra or σ-field

i σ-algebra or σ-field

Let X be some set, and let $\mathcal{P}(X)$ represent its power set. A set $\Sigma \subseteq \mathcal{P}(X)$ is called a σ-algebra or σ-field if and only if it satisfies the following three properties:

1. $X \in \Sigma$.
2. Closed under complementation:
$$A \in \Sigma \implies X \setminus A \in \Sigma.$$
3. Closed under countable unions: For all indexed family $(A_i)_{i\in\mathbb{N}} \subseteq \Sigma$,
$$\bigcup_{i\in\mathbb{N}} A_i \in \Sigma.$$

These imply:

$$\varnothing \in \Sigma$$

Proof.

$$X \in \Sigma \implies X \setminus X = \varnothing \in \Sigma.$$

□

These imply closed under finite unions: For all indexed family $(A_i)_{i\in\mathbb{N}\cap[1,n]} \subseteq \Sigma$,

$$\bigcup_{i=1}^{n} A_i \in \Sigma.$$

Proof. Let $(B_i)_{i\in\mathbb{N}}$ be defined as $B_i = A_i$ for $i \in [1,n]$ and $B_i = \varnothing$ for $i \in (n,\infty)$ and apply closed under countable unions. □

ii Measurable space (可測空間)

A tuple (X,Σ) of a set X and a σ-algebra Σ on X is called a measurable space.

iii Intersection is σ-algebra

Let X be some set, and let $P(X)$ represent its power set. If $\mathcal{F}$ is a set of σ-algebras of X, then $\bigcap_{\Sigma\in\mathcal{F}} \Sigma$ is a σ-algebra of X.

Proof. If $A \in \bigcap_{\Sigma\in\mathcal{F}} \Sigma$, then $\forall\Sigma \in \mathcal{F}: A \in \Sigma$, then $\forall\Sigma \in \mathcal{F}: X \setminus A \in \Sigma$, then $X \setminus A \in \bigcap_{\Sigma\in\mathcal{F}} \Sigma$.

If $\mathcal{S} \subseteq \bigcap_{\Sigma\in\mathcal{F}} \Sigma$ and $|\mathcal{S}| \in \mathbb{N} \cup \{\aleph_0\}$, then $\forall\Sigma \in \mathcal{F}: \mathcal{S} \subseteq \Sigma$, then $\forall\Sigma \in \mathcal{F}: \bigcup_{A\in\mathcal{S}} A \in \Sigma$, then $\bigcup_{A\in\mathcal{S}} A \in \bigcap_{\Sigma\in\mathcal{F}} \Sigma$. □

iv σ-algebra generated by collection of subsets

For a set X and a collection $C \subseteq \mathcal{P}(X)$, the σ-algebra generated by C, denoted as $\sigma(C)$ or $\sigma\{C\}$, is the smallest σ-algebra that is a superset of C.

v Borel σ-algebra

Let X be a topological space, the Borel σ-algebra of it, $\mathcal{B}(X)$, is the σ-algebra generated by the set of all Borel sets of X.

vi Borel measurable space

A tuple $(X, \mathcal{B})$ of a set X and a Borel σ-algebra $\mathcal{B}$ on X is called a Borel measurable space.

vii Intervals generating Borel σ-algebra of the real numbers theorem

The Borel σ-algebra of the real numbers can be generated by the collection of all intervals of the form $(-\infty, x]$ for $x \in \mathbb{R}$.

Proof. For every $a < b$, the interval $(a, b]$ can be expressed as

$$(a, b] = (-\infty, b] \cup (-\infty, a]^c.$$

For every $a < b$, the interval (a, b) can be expressed as

$$(a, b) = \bigcup_{n=1}^{\infty} (a, b - 1/n].$$

□

viii π-system or pi-system

A π-system or pi-system on a set Ω is a set P of subsets of Ω such that

- $P \neq \emptyset$, and
- $A, B \in P \implies A \cap B \in P$.

ix Generated π-system

For any non-empty set Σ of subsets of a set Ω, there exists a π-system P, called π-system generated by Σ, that is the unique smallest π-system of Ω containing every element of Σ. It is equal to the intersection of all π-systems of Ω containing every element of Σ.

Proof. TODO □

x λ-system, lambda-system, Dynkin system, or d-system

A λ-system, lambda-system, Dynkin system, or d-system on a set Ω is a set D of subsets of Ω such that:

- $\Omega \in D$,
- Closed under complements of subsets in supersets: $A, B \in D \wedge A \subseteq B \implies B \setminus A \in D$, and
- Closed under countable increasing union: if $A_1 \subseteq A_2 \subseteq \ldots$ is an increasing sequence of sets in D, then $\bigcup_{i=1}^{\infty} A_n \in D$.

xi π-system, λ-system, and σ-algebra

1. A set is both a π-system and a λ-system iff it is a σ-algebra.
2. A π-system is not necessarily a λ-system.
3. A π-system is not necessarily a σ-algebra.
4. A λ-system is not necessarily a π-system.
5. A λ-system is not necessarily a σ-algebra.

Proof. TODO □

xii (Sierpiński–Dynkin's) π-λtheorem

If P is a π-system and D is a λ-system with $P \subseteq D$, then the σ-algebra generated by P is a subset of D.

Proof. TODO □

II Measure

i Measure (測度)

Let (X, Σ) be a measurable space. A function $\mu : \Sigma \to [-\infty, +\infty]$ is called a (positive) measure on Σ iff it satisfies

- Countable additivity, also called σ-additivity: For all indexed family $(E_k)_{k\in\mathbb{N}}$ of pairwise disjoint sets in Σ,
$$\mu\left(\bigcup_{k\in\mathbb{N}} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$
- Nonnegativity:
$$\forall E \in \Sigma : \mu(E) \geq 0,$$

Countable additivity implies $\mu(\emptyset) = 0$.

Proof. By applying for $(\emptyset)_{k\in\mathbb{N}}$,
$$\mu(\emptyset) = \sum_{k=1}^{\infty} \mu(\emptyset)$$
□

Countable additivity implies finite additivity: For all indexed family $(E_k)_{k\in\mathbb{N}\cap[1,n]}$ of pairwise disjoint sets in Σ,

$$\mu\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} \mu(E_k).$$

Proof. Let $(F_k)_{k\in\mathbb{N}}$ be defined as $F_k = E_k$ for $k \in [1, n]$ and $F_k = \varnothing$ for $k \in (n, \infty)$ and countable additivity. $\square$

If $\mu(E) \geq 0$ is replaced with $\mu(E) \in \mathbb{R}$, it's called a signed measure.

If $\mu(E) \geq 0$ is replaced with $\mu(E) \in \mathbb{C}$, it's called a complex-valued measure.

ii Measure space (測度空間)

A tuple (X, Σ, μ) of a set X, a σ-algebra Σ on X, and a measure μ (sometimes signed) on Σ is called a measurable space.

iii Pre-measure

Let X be a set and $\mathcal{R}$ be ring of subsets on X. A function $\mu : \mathcal{R} \to [-\infty, +\infty]$ is called a signed pre-measure on $\mathcal{R}$ iff it satisfies

- $\mu(\varnothing) = 0$ and
- countable additivity, also called σ-additivity, that, for all finite or countable infinite indexed family $(E_k)_{k\in K}$ of pairwise disjoint sets in $\mathcal{R}$,

$$\mu\left(\bigcup_{k\in K} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

If it also satisfies nonnegativity:

$$\forall E \in \mathcal{R} :\ \mu(E) \geq 0,$$

it is called a pre-measure or positive pre-measure.

iv Outer or exterior measure

Let X be some set, and let $\mathcal{P}(X)$ represent its power set. A function $\mu : \mathcal{P}(X) \to [-\infty, +\infty]$ is called a signed outer or exterior measure on $\mathcal{R}$ iff it satisfies

- $\mu(\varnothing) = 0$ and
- countable subadditivity, that, for all finite or countable infinite indexed family $(E_k)_{k\in K}$ of pairwise disjoint sets in $\mathcal{P}(X)$,

$$A \subseteq \bigcup_{k\in K} E_k \implies \mu(A) \leq \sum_{k=1}^{\infty} \mu(E_k).$$

If it also satisfies nonnegativity:

$$\forall E \in \mathcal{R} :\ \mu_0(E) \geq 0,$$

it is called a outer or exterior measure or outer or exterior measure.

v Product measure space

Let $(X, \mathcal{A}, \mu), (Y, \mathcal{B}, \nu)$ be two measure space with possibly signed measure μ, ν. The sigma algebra of the product measure space $X \times Y$ is

$$\sigma(\{A \times B \mid A \in \mathcal{A} \wedge B \in \mathcal{B}\}),$$

denoted as $\mathcal{A} \otimes \mathcal{B}$, and the possibly signed measure of it is $\mu \times \nu$ restricted to $\mathcal{A} \otimes \mathcal{B}$.

vi σ-finite measure

Let X be a set, $\mathcal{R}$ be a ring of sets on X, and μ being a signed or positive pre-measure. μ is σ-finite iff there exists an index family $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{R}$ such that

$$\forall n \in \mathbb{N}: \ \mu(A_n) < \infty \wedge \bigcup_{n \in \mathbb{N}} A_n = X.$$

If μ is a σ-finite measure, the measure space (X, Σ, μ) is called a σ-finite measure space.

vii Measures agree on generating π-system agree theorem

Suppose μ_1, μ_2 are two measures on a measurable space $(X, \mathcal{F})$, and $\mathcal{F}$ is generated by a π-system P on X. If

$$\forall A \in P: \ \mu_1(A) = \mu_2(A)$$

and

$$\mu_1(X) = \mu_2(X) < \infty,$$

then $\mu_1 = \mu_2$.

Proof. Define $D = \{A \in \mathcal{F}: \ \mu_1(A) = \mu_2(A)\}$, which is a λ-system since $\mu_1(X) = \mu_2(X) < \infty$. By π-λtheorem, $\sigma(P) \subseteq D$. □

viii Carathéodory's extension theorem

Let X be a set, $\mathcal{R}$ be a ring of sets on X, μ_0 be a pre-measure on $\mathcal{R}$, and $\sigma(\mathcal{R})$ be the σ-algebra generated by $\mathcal{R}$.

The Carathéodory's extension theorem states that there exists a measure μ on $\sigma(\mathcal{R})$ such that μ_0 is an extension of μ.

Moreover, if μ_0 is σ-finite, then the extension μ is unique and also σ-finite.

Proof. TODO □

ix Measurable functions

Let (X, Σ) and (Y, T) be two measurable spaces. A function $f: D \subseteq X \to Y$ is said to be measurable iff for every $E \in T$, the preimage

$$f^{-1}(E) \in \Sigma.$$

x Borel measurable functions

Let $(X, \mathcal{B})$ be a Borel measurable space and (Y, T) be a measurable spaces. A function $f : D \subseteq X \to Y$ is said to be Borel measurable iff for every $E \in T$, the preimage

$$f^{-1}(E) \in \mathcal{B}.$$

xi Null set

A set A in the σ-algebra is called a μ-null set for a measure μ if $\mu(A) = 0$.

xii Almost everywhere (a.e.)

If (X, Σ, μ) is a measure space, a property P is said to hold almost everywhere or μ-almost everywhere in $Y \subseteq X$ if there exists a measurable set $N \in \Sigma \wedge N \subseteq Y$ with $\mu(N) = 0$, such that for all $x \in Y \setminus N$, the property P holds at x.

xiii Continuity from below

Let $(A_n)_{n\in\mathbb{N}}$ be an increasing sequence of measurable sets, i.e., $A_{i-1} \subseteq A_i$ for all i, and let $A = \bigcup_{n=1}^{\infty} A_n$. Then a measure μ satisfies

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

Proof. Define

$$B_1 = A_1,$$

$$B_n = A_n \setminus A_{n-1}, \quad n \geq 2.$$

$$\mu(A_n) = \sum_{k=1}^{n} \mu(B_k),$$

$$\mu(A) = \sum_{k=1}^{\infty} \mu(B_k).$$

□

xiv Continuity from above

Let $(A_n)_{n\in\mathbb{N}}$ be a decreasing sequence of measurable sets, i.e., $A_{i-1} \supseteq A_i$ for all i, and let $A = \bigcap_{n=1}^{\infty} A_n$. Then a measure μ such that $\mu(A_1) < \infty$ satisfies

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

Proof. Define

$$B_n = A_1 \setminus A_n.$$

By continuity from below,

$$\mu(\bigcup_{k=1}^{n} B_n) = \mu(A_1 \setminus A) = \lim_{n\to\infty} \mu(A_1 \setminus A_n)$$

Since $A, A_n \subseteq A_1$, by countable additivity,

$$\mu(A_1 \setminus A) = \mu(A_1) - \mu(A)$$

$$\mu(A_1 \setminus A_n) = \mu(A_1) - \mu(A_n)$$

$$\mu(A_1) - \mu(A) = \lim_{n\to\infty} \left(\mu(A_1) - \mu(A_n)\right) = \mu(A_1) - \lim_{n\to\infty} \mu(A_n)$$

Since $\mu(A_1) < \infty$,

$$\mu(A) = \lim_{n\to\infty} \mu(A_n)$$

□

III Borel measure

i Borel measure

A Borel measure is a measure on a Borel σ-algebra.

ii Radon measure

Let X be a locally compact Hausdorff space, and μ be a Borel measure of X.

- μ is called inner regular or tight iff, for every open set U, $\mu(U)$ equals the supremum of $\mu(K)$ over all compact subsets K of U.
- μ is called outer regular iff, for every Borel set B, $\mu(B)$ equals the infimum of $\mu(U)$ over all open sets U that contain B.
- μ is called locally finite iff every point of X has a neighborhood U for which $\mu(U)$ is finite.
- μ is called a Radon measure if it is inner regular and locally finite.

IV Lebesgue measure (勒貝格測度)

i Real numbers

Let R be the set of all open intervals, and $\ell((b-a)) = b-a$ be the length function. For any subset $E \subseteq \mathbb{R}$, the Lebesgue outer measure $\lambda(E)$ is defined as

$$\lambda(E) = \inf\left\{ \sum_{k=1}^{\infty} \ell(I_k) : \ (I_k)_{k\in\mathbb{N}} \subseteq R \wedge E \subseteq \bigcup_{k=1}^{\infty} I_k \right\}.$$

ii Real vector space

Let R be the set of all open rectangle in $\mathbb{R}^n$ with $n \in \mathbb{N}$, that is, $\prod_{i=1}^{n} I_i$ for open intervals I_i. For any $C \in R$, the Lebesgue outer measure $\lambda(C)$ is defined as

$$\lambda(C) = \prod_{i=1}^{n} \ell(I_i).$$

For any subset $E \subseteq \mathbb{R}^n$ with $n \in \mathbb{N}$, the Lebesgue outer measure $\lambda(E)$ is defined as

$$\lambda(E) = \inf\left\{ \sum_{k=1}^{\infty} \lambda(C_k) :\ (C_k)_{k\in\mathbb{N}} \subseteq R \wedge E \subseteq \bigcup_{k=1}^{\infty} C_k \right\}.$$

iii Carathéodory criterion

A set $E \subseteq \mathbb{R}^n$ satisfies the Carathéodory criterion iff

$$\forall A \subseteq \mathbb{R}^n :\ \lambda(A) = \lambda(A \cap E) + \lambda(A \cap (\mathbb{R}^n \setminus E)).$$

A set E that satisfies the Carathéodory criterion is Lebesgue-measurable, with its Lebesgue measure being defined as its Lebesgue outer measure. The set of all such E is the Lebesgue σ-algebra.

$\mathcal{B}(\mathbb{R}^n)$ is a subset of Lebesgue σ-algebra.

A set $E \subseteq \mathbb{R}^n$ that does not satisfy the Carathéodory criterion is not Lebesgue-measurable. ZFC proves that such sets do exist.

Proof. TODO □

iv Unique Borel measure that is translation invariant

Lebesgue measure is the unique Borel measure that is translation invariant, up to scaling by a positive constant.

Proof. TODO □

V Hausdorff measure (郝斯多夫測度)

i Hausdorff outer measure

Let (X, d) be a metric space. For any subset $U \subseteq X$, let $\operatorname{diam}(U)$ denote its diameter, that is, $\operatorname{diam}(\varnothing) = 0$ and for all $U \neq \varnothing$,

$$\operatorname{diam}(U) = \sup\{d(x, y) \mid x, y \in U\}.$$

Let S be any subset of X, and δ be a positive real number. Define

$$H_\delta^d(S) = \inf\left\{ \sum_{k=1}^{\infty} (\operatorname{diam}(U_k))^d \mid S \subseteq \bigcup_{k=1}^{\infty} U_i \subseteq X \wedge \forall k \in \mathbb{N} :\ \operatorname{diam}(U_k) < \delta \right\}.$$

Note that $H^d_\delta(S)$ is nonincreasing in δ. Thus, $\lim_{\delta\to 0} H^d_\delta(S) \in [0,\infty]$. The Hausdorff outer measure $H^d(S)$ is defined as

$$H^d(S) := \lim_{\delta\to 0} H^d_\delta(S).$$

ii Carathéodory criterion

A set $E \subseteq X$ satisfies the Carathéodory criterion iff

$$\forall A \subseteq X :\ H^d(A) = H^d(A \cap E) + H^d(A \cap (X \setminus E)).$$

A set E that satisfies the Carathéodory criterion is Hausdorff-measurable, with its Hausdorff measure being defined as its Hausdorff outer measure. The set of all such E is the Hausdorff σ-algebra.

$\mathcal{B}(X)$ is a subset of Hausdorff σ-algebra.

Proof. TODO □

A set $E \subseteq X$ that does not satisfy the Carathéodory criterion is not Hausdorff-measurable.

VI Hahn decomposition theorem

i Hahn decomposition theorem

TODO

ii Jordan decomposition theorem

TODO

VII Radon–Nikodym derivative

i Absolute continuity (AC) of measures

Consider a measurable space (X,Σ) on which two measures μ,ν are defined. If for every ν-measurable set A, $\nu(A) = 0$ implies $\mu(A) = 0$, equivalently $\mu(A) > 0$ implies $\nu(A) > 0$, then we write $\mu \ll \nu$ and say μ is dominated by ν or absolutely continuous w.r.t. ν.

ii Radon–Nikodym theorem and derivative

Consider a measurable space (X,Σ) on which two σ-finite measures μ,ν are defined. If $\nu \ll \mu$, then there exists a Σ-measurable function $X \to [0,\infty)$ such that for any measurable set $A \in \Sigma$,

$$\nu(A) = \int_A f \,\mathrm{d}\mu\,.$$

If the function f exists, it us uniquely defined up to a μ-null set, that is, if g is another function satisfying the same property, then $f = g$ μ-a.e. The function f is commonly written $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ and is called Radon–Nikodym derivative.

Proof. TODO □

iii Linearity

If $\nu, \mu \ll \lambda$ and $a, b \in \mathbb{R}_{\geq 0}$, then

$$\frac{\mathrm{d}(a\nu + b\mu)}{\mathrm{d}\lambda} = a\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} + b\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}$$

λ-a.e.

Proof. TODO ☐

iv Chain rule

If $\nu \ll \mu \ll \lambda$, then

$$\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}$$

λ-a.e.

Proof. TODO ☐

v Reciprocal rule

If $\mu \ll \nu$ and $\nu \ll \mu$, then

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu} = (\frac{\mathrm{d}\nu}{\mathrm{d}\mu})^{-1}$$

ν-a.e.

Proof. TODO ☐

vi Integral by substitution

If $\mu \ll \lambda$ and $g \in L^1(X, \mu)$, then

$$\int_X g \,\mathrm{d}\mu = \int_X g\frac{\mathrm{d}\mu}{\mathrm{d}\lambda} \,\mathrm{d}\lambda\,.$$

Proof. TODO ☐